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Empirical and machine learning-based approaches to identify rainfall thresholds for landslide prediction: a case study of Kerala, India

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Abstract
For the safe and resilient development of cities, disaster risk reduction plays a crucial role, aligning with sustainable development goal 11 of the United Nations. Supporting this objective, the present study developed a machine learning (ML) classifier-based threshold model to determine rainfall thresholds for predicting impending landslides in Kerala, India, using historical data. Using a dataset of 64 rainfall-induced landslide events recorded since the year 2000, rainfall data were collected up to 15 days prior to each landslide to support empirical analysis of intensity-duration and event rainfall-duration thresholds. In cases where exact rainfall durations were unavailable, classification machine learning (ML) models, including K-nearest neighbours (KNN), random forest (RF), gradient boosting machine (GBM), support vector machine (SVM), and logistic regression, were used to determine threshold reliability. Among these, the KNN model with 5 Neighbours achieved the highest performance, with an ROC-AUC of 0.9 and an accuracy of 82%. This model, saved as a pickle file, serves as a core filter in the development of a landslide early warning system. This paper presents the model development and performance comparisons, contributing to a practical, community-centred solution for landslide disaster resilience in Kerala.

Highlights

- Kerala, a state in India, experiences one of the highest incidences of rainfall-induced landslides. Historical data has been collected and analyzed to devise thresholds for the early detection of landslides.
- Two empirical approaches based on the relationships between rainfall intensity and duration, as well as cumulative rainfall and duration, have been utilized to identify early warning thresholds for landslides.
- Five machine learning-based approaches were employed to determine these thresholds. Among the classifiers tested, the K-Nearest Neighbour (KNN) classifier with K=5 demonstrated the highest prediction accuracy compared to other methods in the study.

Keywords Rainfall-induced landslide · Rainfall threshold · Machine Learning · Empirical models

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1 Introduction

Disaster risk reduction is a fundamental component of the sustainable development of cities. The United Nations' sustainable development goals (SDG) emphasize this as an integral aspect of Goal 11, which seeks to make cities inclusive, safe, resilient, and sustainable [1]. This goal highlights the importance of reducing the risks and impacts of disasters to ensure long-term urban resilience and sustainability. One effective way to support disaster risk reduction is through the development and implementation of policies designed to make cities more resilient and better prepared to face impending disasters. Resilient cities are those equipped with the knowledge, infrastructure, and systems necessary to mitigate disaster impacts and recover effectively. For disasters caused due to landslides, achieving such resilience requires a comprehensive understanding of their mechanisms and initiation processes. A critical step in addressing landslide risks is to the study and analysis of past events to identify key contributing factors. Examining historical landslide behavior provides invaluable insights into the causes, triggering factors, and threshold parameters associated with such events. This data can guide the development of predictive models and early warning systems, enabling timely and informed decision-making. By understanding the interaction between environmental conditions, human activities, and geological factors, it becomes possible to devise strategies that minimize risk and enhance preparedness, contributing significantly to the goal of sustainable urban development.

Kerala, a southwestern coastal state of India, frequently experiences severe landslides during heavy rains. This study focuses on analyzing the frequency and distribution of these landslides in context of varying rainfall pattern. Kerala is geographically situated between latitudes 8° 17′ 30″ N to 12° 47′ 40″ N and longitudes 74° 27′ 47″ E to 77° 37′ 12″ E. The state's eastern boundary is marked by the Western Ghats, which rise to approximately 2000 m above sea level, with peaks reaching up to 2500 m. On the western side, Kerala is bordered by the Arabian Sea. The state's terrain is undulating, with a total area of 38,863 km^2 and 44 rivers flowing through it. The Western Ghats, never more than 120 km from the sea, cover 47% of Kerala, making the region highly prone to landslides. Kerala experiences two major rainy seasons: the Southwest Monsoon, which begins in June, and the Northeast Monsoon, starting in mid-October. These heavy monsoon rains often trigger landslides, especially in the hilly regions of the Western Ghats. Thirteen of Kerala's fourteen districts, excluding the coastal district of Alappuzha, are affected by landslides annually. The state's high annual rainfall contributes to the occurrence of numerous landslides each year [2, 3]. Figure 1 shows the past landslide recorded by the authorities in the state of Kerala.

India's Western Ghats are widely recognized as a major region prone to rainfall-triggered landslides, accounting for over 59% of India's landslide occurrences. [4, 5]. While natural factors like rainfall patterns and topography are beyond human control, landslides stem from the Earth's surface adjusting to shifting atmospheric conditions. These incidents only become disasters when they threaten human life or infrastructure, highlighting the human element in landslide catastrophes. Traditionally, local communities observed warning signs of potential slope instability, such as ground cracks, tilting trees, and unusual animal behavior, to predict landslides. [6]. Despite technological advancements, preventing landslide disasters remains difficult due to several challenges. Notably, frequent false alarms can lead to the residents often ignoring warnings. This emphasizes the need for an affordable, community-based early warning system that provides household-specific alerts for those in high-risk landslide areas. Such a system would equip communities with timely, dependable warnings, fostering proactive mitigation efforts and enhancing resilience in landslide-prone regions.

Various types of landslide early warning systems (LEWS) have been developed over the years, with initial studies focusing on the relationship between rainfall intensity and duration (ID) thresholds. A seminal study by Caine (1980) established a critical threshold for rainfall intensity (measured in mm/hr) in relation to duration (in hours) based on data from 73 landslide events worldwide [7, 8]. While foundational, this method faces limitations, particularly in regions with prolonged rainfall patterns like the Western Ghats, where continuous precipitation over days or weeks disrupts the ID model's accuracy. This approach struggles to differentiate landslides triggered by a single day of rainfall from those caused by cumulative precipitation over extended periods, given that short dry intervals do not allow the soil to recover sufficiently. To address these challenges, researchers have explored alternative event rainfall-duration (ED) thresholds, which consider cumulative rainfall over a specified period. This method is especially beneficial in regions where rainfall accumulates without allowing time for the soil to regain stability. These cumulative thresholds aim to provide a more realistic assessment of landslide risk under prolonged rainfall conditions. Another extensively researched approach for landslide prediction is direct monitoring through on-site instrumentation, such as soil moisture sensors and tilt meters. Although effective in capturing local soil and slope conditions, these systems

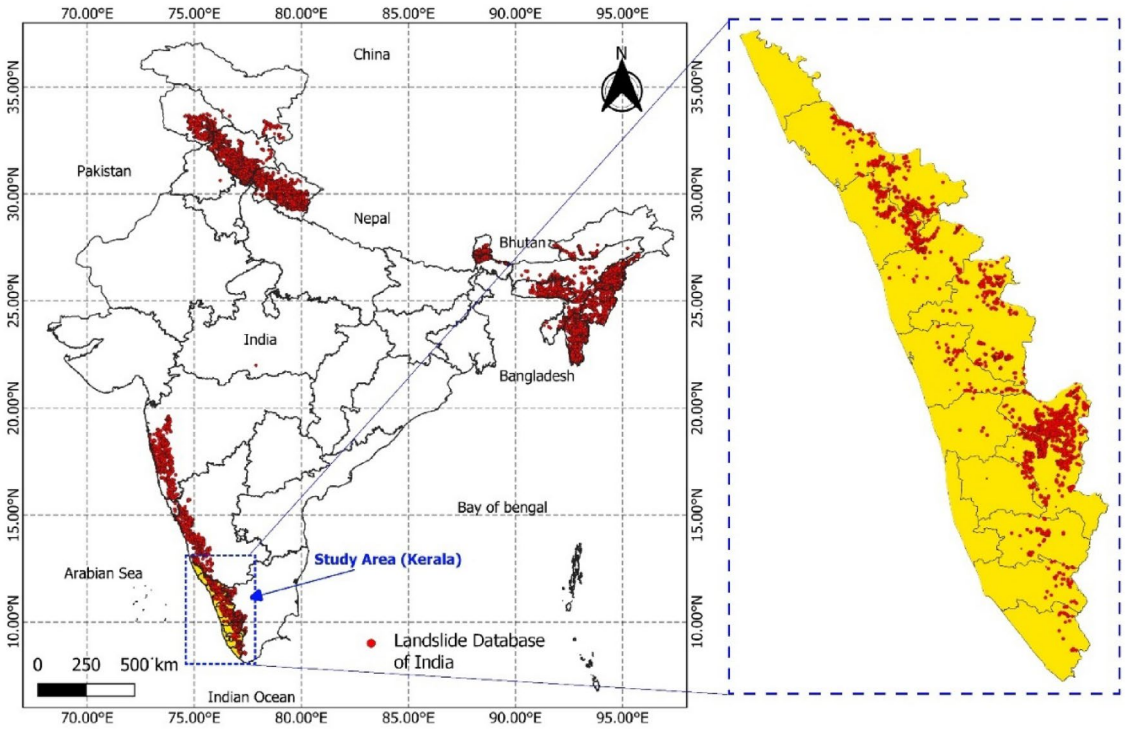


Fig. 1 Spatial distribution of 3164 landslides of Kerala from the geological survey of India (GSI) archive [3]

are inherently limited by their scope; monitoring is generally restricted to instrumented slopes, leaving unmonitored areas susceptible to undetected landslides. Moreover, installing a vast network of sensors across expansive landslide-prone areas like the Western Ghats is financially challenging and logistically complex [9]. To overcome some of these limitations, remote sensing has become a valuable tool for landslide monitoring. Techniques such as interferometric synthetic aperture radar (InSAR) allow for satellite-based mapping of surface deformations indicative of slope instability [10]. While this method offers broader spatial coverage, it relies on frequent satellite passes to gather real-time data and may not provide warnings quickly enough to alert the public before a landslide occurs. A more recent advancement in LEWS is the integration of Internet of Things (IoT) sensors with advanced communication technologies. IoT-based systems provide real-time data transmission and enable direct communication with residents in high-risk areas. These systems leverage internet connectivity to monitor slope conditions continuously, allowing for rapid dissemination of alerts to vulnerable communities [11, 12]. This approach is increasingly preferred due to its affordability, scalability, and potential to deliver location-specific, timely warnings, making it a promising solution for improving resilience in landslide-prone regions.

Machine learning (ML) techniques have proven valuable in predicting specific events based on historical data. [13–15]. This capability extends to forecasting natural disasters like landslides, where ML models can analyze patterns and trends to identify potential triggers. Numerous studies have applied ML techniques to remote-sensing data [16–18] and to determine rainfall thresholds [19–21] that signals landslide risks. The quality and relevance of data are critical to the effectiveness of any ML model. For landslide prediction, rainfall data up to 2023 is sourced from the Indian meteorological department's (IMD) archives, where open-source gridded data provides approximate rainfall measurements for specific sites across India. Over the past 20 years, rainfall data from sites with recorded landslide events were compiled, covering each landslide occurrence and the rainfall in the preceding 15 days. This approach acknowledges that landslides are typically cumulative events, not merely triggered by a single day of rainfall. [22–24]. The data structure includes a binary classification target, where each data point is labelled as either "Landslide" or "No Landslide." This binary outcome is crucial, as it frames the problem for the ML model, which will use historical patterns to distinguish between conditions likely to cause landslides and those that are not. Figure 2 illustrates various data points collected over the selected period and locations.

This research utilizes a unique 24-year dataset (2000–2024) of landslide occurrences in Kerala to develop a robust rainfall threshold model. This dataset (Fig. 2a), which has not been used in any previous studies, serves as the foundation for both

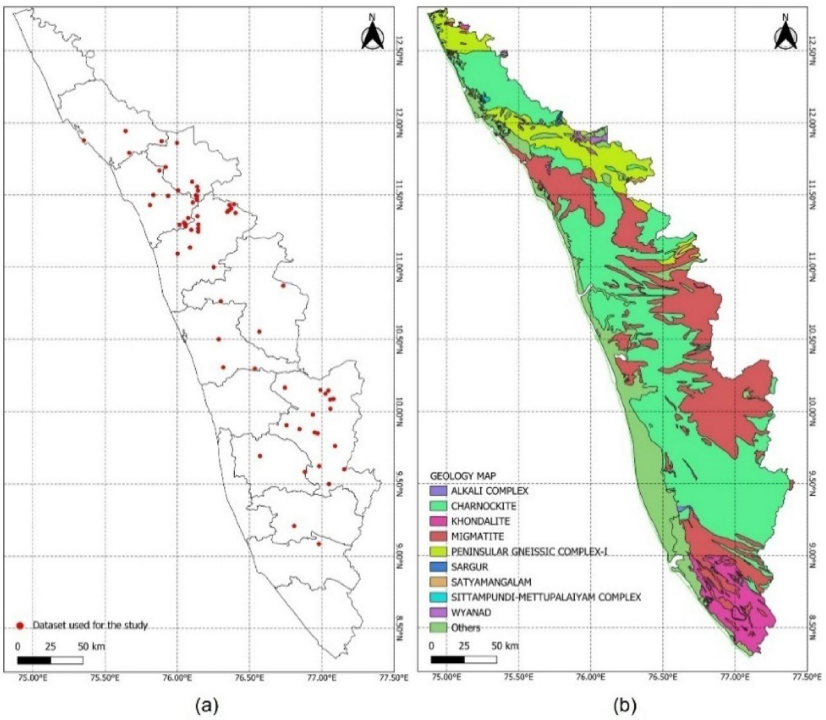


Fig. 2 **a** 64 landslide points used for the present study and **b** geological map of Kerala

empirical and machine learning approaches. The distinctive characteristics of the dataset play a crucial role in shaping the model's fit and predictive performance, ensuring region-specific insights into landslide prediction. Moreover, the selected data points highlight the geological composition of the study area (Fig. 2b), which is predominantly characterized by charnockite and migmatite formations. These geological features are integral to understanding the soil's behavior and its susceptibility to rainfall-induced landslides, further reinforcing the validity and applicability of the developed models. Previous literature has predominantly relied on empirical models and limited machine learning techniques: Naidu et al. applied a simple regression-based empirical model [25], Abraham et al. employed an ID threshold based on Caine's 1980 framework [26], and Sheelu et al. explored Artificial Neural Networks and logistic regression [17]. Thus, there is a clear gap in the use of advanced machine learning classifiers, highlighting the novelty and potential contribution of this study.

Machine learning (ML) techniques such as logistic regression (LR), support vector machine (SVM), gradient boosting method (GBM), random forest (RF), and K-nearest neighbour (KNN) have been widely utilized by previous researchers for the development of landslide susceptibility mapping and the establishment of regional prediction thresholds [27–31]. These algorithms are particularly popular because they are capable of handling complex, nonlinear relationships within datasets, enabling more accurate predictions compared to traditional methods. Given their proven effectiveness, these popular ML techniques have been tested and evaluated in this study against traditional methods, specifically the intensity-duration (ID) and event-duration (ED) thresholds, to benchmark their performance and highlight potential improvements in predictive accuracy for landslide forecasting.

The present study utilizes the data provided in Fig. 2 and Fig. 3, which has been extracted from the landslide inventory data given in Fig. 1 and IMD database. The data will be used to develop empirical models and machine learning models, especially intensity-duration thresholds, rainfall event duration thresholds, K-nearest neighbour (KNN) classifier, random forest (RF) classifier, support vector machine (SVM) classifier, logistic regression (LR) classifier, gradient boost machine (GBM) classifier.

2 Methodology

2.1 Empirical methods

In previous studies, rainfall thresholds have been determined using empirical methods to identify conditions that are likely to trigger landslides. One of the most widely used approaches is the development of intensity-duration (ID)

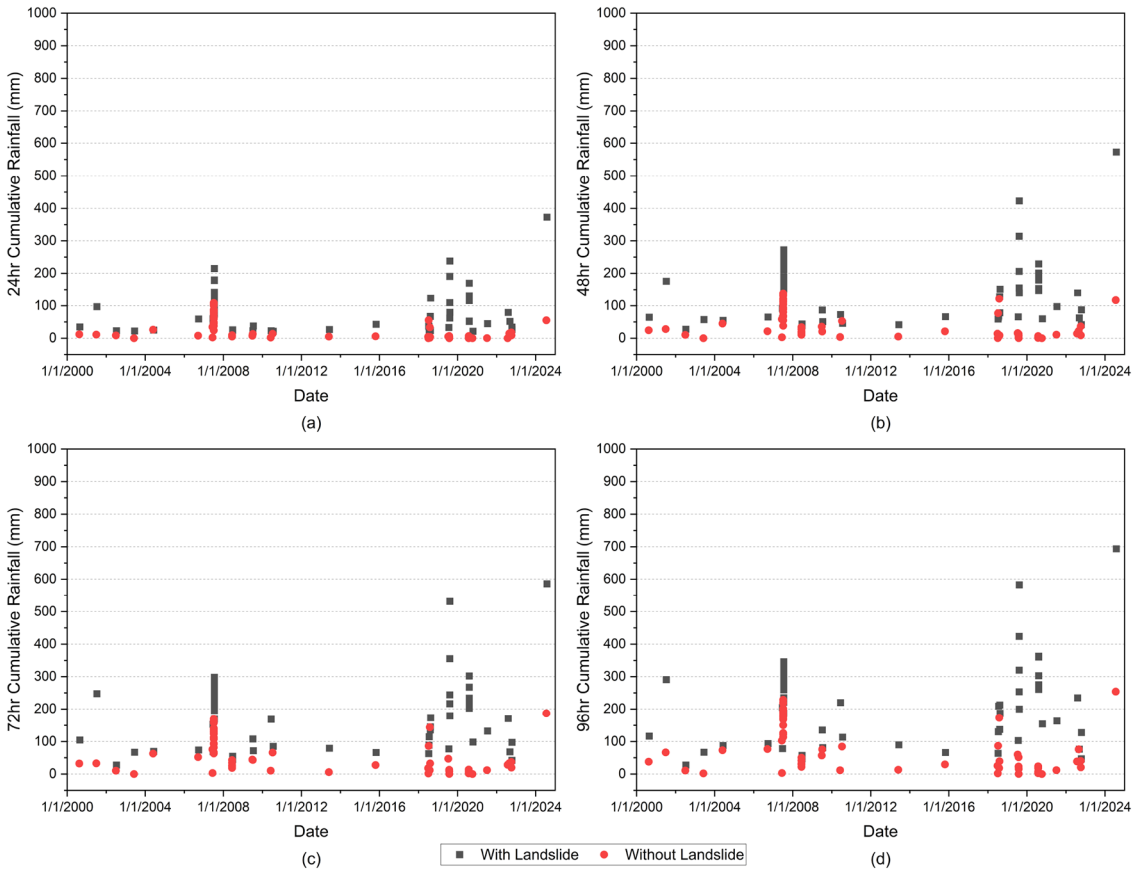


Fig. 3 Cumulative rainfall data for **a** 24 h, **b** 48 h, **c** 72 h, **d** 96 h

thresholds, which quantify the relationship between rainfall intensity and duration as a predictive measure. The general form of an ID threshold, as described in Eq. 1, was first introduced by Caine [7],

$$I = C + \alpha D^{-\beta} \tag{1}$$

where I represent the rainfall intensity in mm/hour, D is the duration of rainfall in hours, α and β are fitting parameters specific to the region and study, and C is a constant greater than or equal to zero. This threshold helps determine the critical point where the intensity and duration of rainfall are likely to initiate landslide activity. Another commonly used threshold is the event-duration (ED) threshold, which focuses on the cumulative rainfall over a given period. As shown in Eq. 2, the ED threshold defines a threshold based on the total rainfall accumulated over a specific duration. [7, 32],

$$E = \alpha D^{\beta} \tag{2}$$

In this equation, E is the cumulative rainfall in mm, D is the duration in hours, and α and β are fitting parameters derived from historical data. These parameters are essential as they allow for customization of the model to reflect local or regional rainfall patterns associated with landslide events. Both ID and ED thresholds serve as important tools in landslide early warning systems by identifying rainfall conditions that correlate with landslide occurrences. By applying these thresholds to historical and real-time rainfall data, researchers and authorities can assess landslide risks and improve disaster preparedness. Figure 4 shows the intensity duration and rainfall event duration plots used for the present study.

2.2 Machine learning models

In this study, several machine learning classifiers were employed to predict landslide occurrences based on historical rainfall data, providing a robust analysis of factors contributing to landslide events. These classifiers included the K-nearest neighbors (KNN), random forest (RF), logistic regression, gradient boosting machine (GBM), and support vector machine

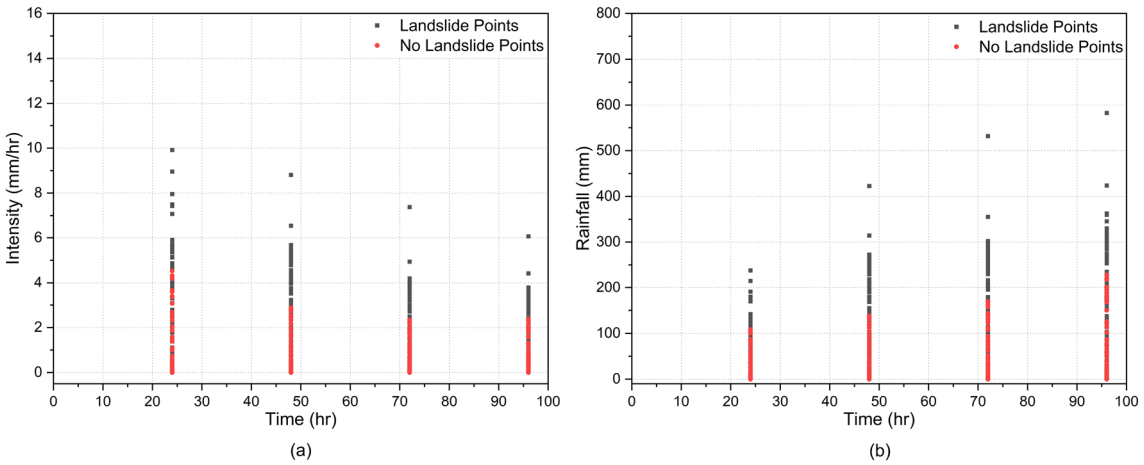


Fig. 4 **a** Rainfall intensity vs Time **b** Rainfall event vs Time

(SVM) classifiers, each selected for their unique strengths in handling binary classification tasks. In this study, the KNN classifier was trained on a dataset containing 64 instances of landslide events and 64 instances of non-landslide events, ensuring a balanced dataset for optimal classification accuracy. The landslide and non-landslide data were collected from the GSI and IMD. [4, 33, 34], and the dataset's summary statistics are shown in Table 1.

The KNN algorithm is a widely used supervised learning method known for its effectiveness in classification and regression. In KNN, the parameter K denotes the number of nearest data points (neighbors) that contribute to classifying an instance within a specific region of the dataset. KNN does not rely on assumptions about the data distribution, making it particularly useful for binary data. This method requires both positive and negative instances to function accurately. The Random Forest classifier, an ensemble method, builds multiple decision trees and aggregates their outputs to enhance classification accuracy and reduce overfitting. By combining the results of several decision trees, RF captures complex patterns within the data. This classifier is particularly useful for handling imbalanced data and noisy features, as it naturally selects relevant variables for classification. In this study, RF was used to model landslide probability based on binary input data, where rainfall intensity and duration from landslide and non-landslide events were analyzed. Logistic Regression, a statistical method for binary classification, was also applied to assess landslide risks. It calculates the probability of a binary outcome by fitting a logistic function to the data. Logistic Regression is advantageous for binary classification problems as it directly outputs probabilities, making it well-suited for distinguishing between landslide and non-landslide events. In this study, it was utilized to interpret the relationship between rainfall variables and the likelihood of landslide occurrence. The Gradient Boosting Machine classifier, an ensemble technique, constructs a series of decision trees that progressively improve the model's predictive accuracy by minimizing errors from previous iterations. GBM is highly effective in capturing complex non-linear patterns in the data, making it useful for landslide prediction where multiple contributing factors interact. For this study, GBM analyzed rainfall data to model landslide probability and distinguished landslide-prone conditions. The Support

Table 1 Data statistics for the rainfall threshold

Parameters	Time (hr)	Rainfall (mm)	Intensity (mm/hr)	Landslide (1/0)
Count	512	512	512	512
mean	60	101.48	1.83	0.5
Standard deviation	26.86	96.54	1.72	0.5
minimum	24	0	0	0
25%	42	24.36	0.45	0
50%	60	69.08	1.34	0.5
75%	78	153.74	2.79	1
Maximum	96	582.92	9.92	1
Pearson Correlation	-2.75e-16	0.51	0.52	1

Vector Machine classifier works by finding a hyperplane that best separates data points into binary classes. SVM is especially effective in high-dimensional spaces and is robust to outliers, making it suitable for analyzing rainfall conditions and other environmental factors that contribute to landslides. In this study, the SVM classifier was employed to differentiate between landslide and non-landslide instances based on rainfall patterns.

2.3 Data preparation

Figure 5 shows the process involved in identifying the data points for the present study. The dataset used in this study consisted of binary classifications, where a value of 1 indicated a landslide and 0 indicated no landslide. Landslide events (64 instances) included rainfall data from the day of the landslide event and three days prior, forming a cumulative 96-h dataset for each event. Similarly, non-landslide data (64 instances) were compiled from the same locations, with rainfall data covering the period 10 to 13 days before each landslide event. This careful selection ensures a balanced dataset for effective training and validation, allowing the models to differentiate between landslide-prone and non-landslide conditions accurately. This balanced dataset provides a robust foundation for training and validating the empirical and machine-learning models used in this study. By comparing landslide and non-landslide conditions under similar environmental factors, these models offer insights into threshold rainfall patterns associated with landslide risks, enabling more accurate and early warnings for disaster preparedness. Figure 6 shows the overall process involved in this methodology.

The best-performing model, identified through rigorous evaluation, will be utilized to develop an economically viable early warning system for rainfall-induced landslides in Kerala, India. To enhance the dataset, the initial 64 landslide records have been expanded to 512 by incorporating both landslide and non-landslide points. This was achieved by considering up to 96 h of rainfall data, which includes the rainfall amounts on the day of the landslide occurrence, the cumulative rainfall for the 96 h leading up to the event, and rainfall data from the same location during the 10 days preceding the landslide, ensuring that periods without landslides are adequately represented. This comprehensive approach allows for a more robust dataset that captures the relationship between rainfall patterns and landslide occurrences. The modified dataset will be randomly divided into training and testing sets at a 70/30 ratio. [35]. This allocation ensures that the model is well-trained on the majority of the data while retaining a substantial portion for testing its predictive capabilities. By implementing this methodology, the early warning system aims to provide timely alerts to mitigate the risks associated with landslides, ultimately enhancing public safety and preparedness in the region.

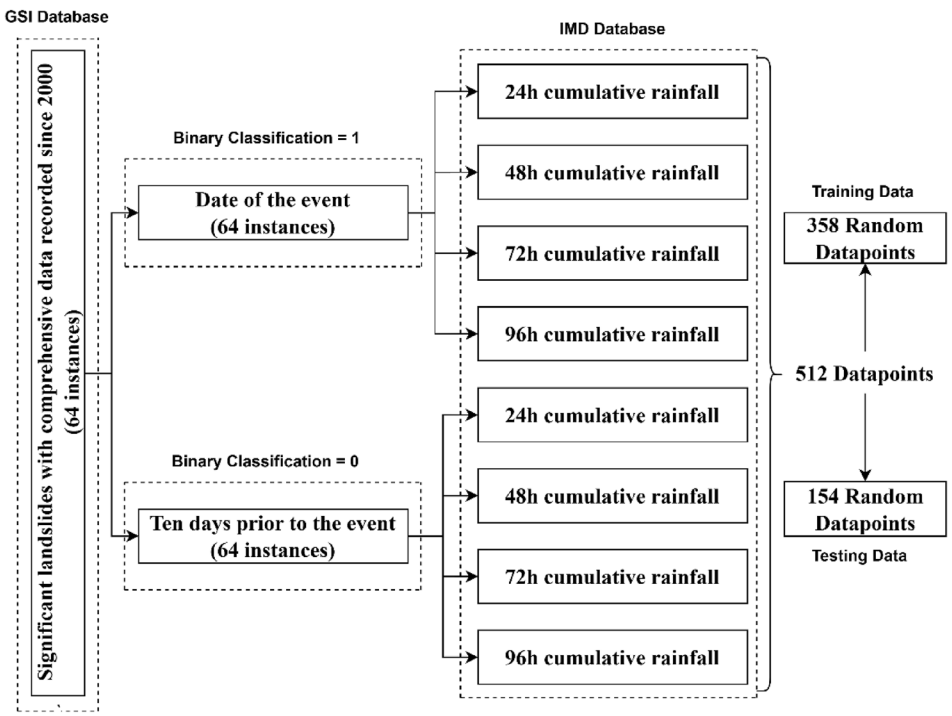


Fig. 5 Flow chart of the methodology of selection of the data points for the study

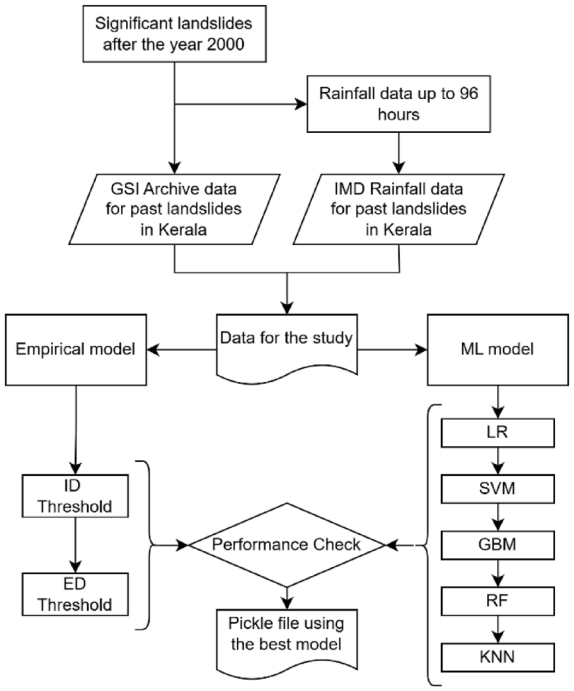


Fig. 6 Methodology chart

2.4 Performance analysis of models

The accuracy of the prediction models devised from the present study, which uses the data set which includes the rainfall data of the landslides has been done differently for different models. In empirical models, mean absolute error (MAE), mean squared error (MSE), and R-squared (R^2) are key metrics for evaluating model performance. MAE measures the average magnitude of prediction errors without considering their direction, providing a straightforward interpretation in the same units as the target variable. MSE, on the other hand, calculates the average of squared differences between predicted and actual values, emphasizing larger errors and making it sensitive to outliers. R-squared (R^2) indicates the proportion of variance in the dependent variable explained by the independent variables, with values ranging from 0 to 1; a higher R^2 signifies a better model fit. Together, these metrics offer valuable insights into the accuracy and effectiveness of regression models. [13, 36]. Equation 3 to 5 shows the equations corresponding to each of these performance indices, where a_i is the ith actual value p_i is the ith predicted value, n is the total number of dataset and $\overline{a}$ is the mean of actual values.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|a_i - p_i| \tag{1}$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(a_i - p_i)^2 \tag{2}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(a_i - p_i)^2}{\sum_{i=1}^{n}(a_i - \overline{a})^2} \tag{3}$$

In classification machine learning models, performance evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix are essential for understanding the model's effectiveness in distinguishing between classes especially when the target parameter is binary. Accuracy measures the overall correctness by calculating the ratio of correctly predicted instances (both true positives and true negatives) to the total number of instances. Precision quantifies the model's ability to identify only relevant instances among the predicted positives, calculated as the ratio of true positives to the sum of true positives and false positives. High precision indicates that positive predictions are typically correct.

Recall assesses the model's ability to identify all relevant instances from the actual positives, calculated as the ratio of true positives to the sum of true positives and false negatives. High recall suggests effective capture of actual positive instances. The F1-score is the harmonic mean of precision and recall, providing a balanced metric that accounts for both false positives and false negatives, making it especially useful in imbalanced datasets. Lastly, the confusion matrix offers a tabular representation of the model's predictions compared to actual classes, displaying counts of true positives, true negatives, false positives, and false negatives, which helps visualize the model's performance and identify error areas. Together, these metrics provide a comprehensive evaluation of classification models, enabling informed decisions about potential improvements. [37–39]. Equation 6 to 9 shows the equations corresponding to Accuracy, Precision, Recall and F1 score. Where, T^+ is the number of true positives, T^- is the number of true negatives, F^+ is the number of false positives and F^- is the number of false negatives.

$$Accuracy = \frac{T^+ + T^-}{T^+ + T^- + F^+ + F^-} \tag{4}$$

$$Precision = \frac{T^+}{T^+ + F^+} \tag{5}$$

$$Recall = \frac{T^+}{T^+ + F^-} \tag{6}$$

$$F1score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{7}$$

3 Results and discussion

The collected data were subjected to a comprehensive evaluation using various mathematical and statistical models to determine the most effective approach for predicting landslide occurrences. The models employed in this analysis include the ID threshold model, ED threshold model, logistic regression model, support vector machine model, gradient boosting model, random forest classifier, and K-nearest neighbor classifier. For the KNN, logistic regression, support vector machine, gradient boosting, and random forest models, the input values used are the intensity and duration data collected from the IMD archive. In contrast, the ED threshold model utilized cumulative rainfall and duration as input variables. In the subsequent sections, we will discuss the results obtained from each of these models in detail. Notably, the analysis reveals that the KNN classifier outperformed all other models, achieving a significantly high accuracy rate in predicting landslide occurrences. This performance suggests that KNN is particularly well-suited for the dataset and the specific context of this study, making it a valuable tool for early warning systems in landslide-prone areas.

3.1 ID threshold

The fitting was performed by using python programming language on the equation developed by Caine (1980). The fitted parameters from the intensity-duration threshold model indicate a baseline intercept (C) of zero, suggesting no landslide risk when the duration is zero. The parameter α, valued at 9.4962 rounded to 9.5, shows that intensity decreases rapidly with increasing duration, while β, at 0.3172, suggests a diminishing return in intensity reduction relative to duration. Model performance metrics reveal a mean absolute error (MAE) of 1.4506, indicating an average prediction deviation, and a mean squared error (MSE) of 3.0592, highlighting significant prediction errors. Notably, the R-squared (R^2) value of 0.0683 indicates that only about 6.83% of the variance in intensity is explained by the model, suggesting a weak fit. These results imply that while the model provides some insights into the intensity-duration relationship, it may require refinement through data quality checks, exploring more complex modeling approaches, incorporating additional relevant variables, and further evaluation techniques to enhance its predictive capability.

Figure 7 illustrates the intensity-time threshold relationship for landslides, depicting both the observed data and the fitted curve derived from the model. The blue dots represent the observed intensity values at various time intervals (duration), while the red curve illustrates the fitted relationship defined by the equation $I = 0 + 9.5D^{-0.32}$.

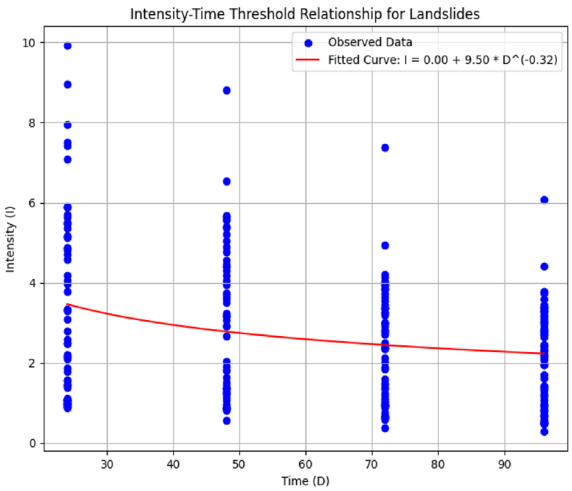


Fig. 7 Fitted curve for the intensity-duration threshold

The fitted curve suggests an inverse relationship between intensity (I) and time (D), indicating that as the duration increases, the intensity tends to decrease. This relationship highlights the model's attempt to capture the threshold conditions under which landslides may occur, demonstrating the potential predictability of landslide intensity based on historical data. However, the spread of the observed data points, particularly at lower intensity levels, indicates variability that may not be fully accounted for by the model, as reflected in the statistical metrics obtained.

3.2 ED threshold

The results from the rainfall-duration model fitting indicate a strong relationship between rainfall intensity (E) and duration (D). The fitted parameters are $\alpha = 12.3216$ rounded to 12.31 and $\beta = 0.6184$ rounded to 0.62, suggesting that the rainfall increases as a power function of the duration. Specifically, this relationship can be described by the equation $E = 12.32D^{0.62}$. The model's performance is quantified by the mean absolute error (MAE) of 76.9114 and a Mean Squared Error (MSE) of 8598.7401, indicating a moderate level of error in the predictions. The R-squared value of 0.1915 reveals that approximately 19.15% of the variance in rainfall can be explained by the duration, suggesting that while there is a correlation, additional factors may influence rainfall intensity. Figure 8 illustrates the observed data points, represented by blue dots, alongside the fitted curve, which provides a visual representation of the model's predictive capability. This plot serves to highlight the trend and relationship between rainfall and duration, further emphasizing the need for continued investigation into the complexities of this relationship in the context of landslide susceptibility.

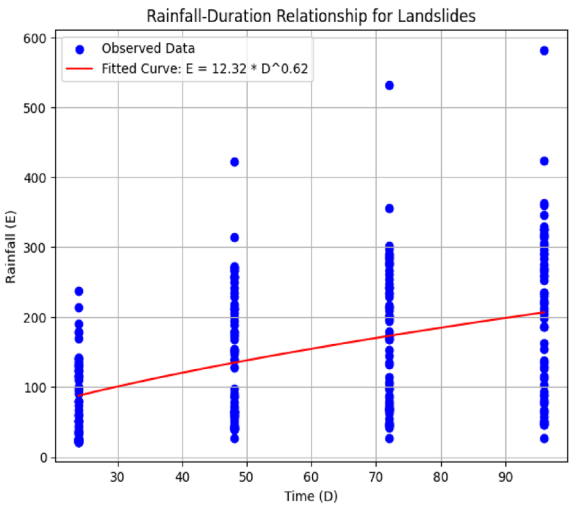


Fig. 8 Fitted curve obtained from the relation between event rainfall and duration

3.3 Logistic regression model

The results from the logistic regression (LR) model provide a comprehensive evaluation of its performance in predicting outcomes. The model achieved an accuracy of 69.48%, indicating that it correctly classified approximately 69.5% of the instances in the dataset. Precision, which reflects the proportion of true positive results among all positive predictions, was reported at 0.75, suggesting that 75% of the positive identifications made by the model were accurate. The recall, or sensitivity, was calculated at 0.63, meaning the model correctly identified 63% of all actual positive cases. The F1-Score, which balances precision and recall, stood at 0.69, demonstrating a reasonable level of model performance given the trade-off between the two metrics. The confusion matrix of the model's classification performance shows 55 true positives, 17 false positives, 30 false negatives, and 52 true negatives. This matrix indicates some challenges in correctly identifying positive cases, as evidenced by the 30 false negatives. Additionally, the receiver operating characteristic area under the curve (ROC-AUC) score of 0.84 signifies a strong ability of the model to distinguish between the positive and negative classes. The ROC-AUC score of 0.84 reinforces the model's effectiveness, suggesting that it has a good overall performance in classification tasks related to landslide susceptibility.

3.4 Gradient boost model

The gradient boosting model (GBM) developed for predicting landslide occurrences based on rainfall intensity and duration has shown promising results, achieving an overall accuracy of 76.62%. This indicates that the model correctly identifies landslide and non-landslide points approximately three-quarters of the time. The key metrics reveal a precision of 0.78, meaning that when the model predicts a point as a landslide, it is correct 78% of the time. Similarly, the recall of 0.78 signifies that the model successfully identifies 78% of the actual landslide instances. The F1-Score, which is also 0.78, reflects a balanced performance between precision and recall. The confusion matrix highlights the model's performance: out of 100 predictions, it correctly classifies 64 true landslides (true positives) and 54 non-landslides (true negatives) while misclassifying 18 non-landslide points as landslides (false positives) and failing to identify 18 actual landslides (false negatives). Furthermore, the model's discriminative ability is confirmed by an ROC-AUC score of 0.89, indicating its effectiveness in distinguishing between landslide and non-landslide occurrences. Overall, the gradient boosting model demonstrates robust predictive performance, showcasing its potential utility in landslide risk assessment and early warning systems. Future research could focus on refining the model by incorporating additional features or optimizing hyperparameters to enhance its predictive capabilities further.

3.5 Support vector machine

The support vector machine (SVM) classifier developed for predicting landslide occurrences based on rainfall intensity and duration has demonstrated varying performance, achieving an overall accuracy of 49.35%. This relatively low accuracy indicates that the model struggles to differentiate between landslide and non-landslide points effectively. However, the precision is exceptionally high at 1.00, suggesting that when the model predicts a point as a landslide, it is correct 100% of the time. In contrast, the recall is quite low at 0.05, indicating that the model only identifies 5% of the actual landslide instances. The F1-Score, calculated at 0.09, reflects the imbalance between precision and recall. The confusion matrix further illustrates this performance: the model correctly classifies 72 true non-landslides (true negatives) but fails to identify 78 actual landslides (false negatives) while also predicting 0 false positives. The model's ability to discriminate between classes is indicated by the ROC-AUC score of 0.83, which confirms its potential for distinguishing landslide occurrences, albeit with challenges in sensitivity. Overall, while the SVM classifier exhibits strong precision, the low recall and accuracy suggest that further refinement is necessary to improve its predictive performance in landslide risk assessment. Future efforts could focus on optimizing model parameters or incorporating additional features to enhance sensitivity and overall effectiveness.

3.6 Random forest classifier

The random forest classifier developed for predicting landslide occurrences based on rainfall intensity and duration has shown strong performance, achieving an overall accuracy of 79.22%. This indicates that the model effectively

distinguishes between landslide and non-landslide points in a majority of cases. The precision of 0.83 suggests that when the model predicts a landslide, it is correct 83% of the time, reflecting a high level of reliability in its positive predictions. The recall of 0.77 indicates that the model successfully identifies 77% of actual landslide instances, demonstrating its effectiveness in capturing relevant events. The F1-score, calculated at 0.80, provides a balanced view of precision and recall, indicating good overall performance. The confusion matrix illustrates the model's performance: it correctly classifies 63 true landslides (true positives) and 59 true non-landslides (true negatives) while misclassifying 19 actual landslides as non-landslides (false negatives) and predicting 13 non-landslide points as landslides (false positives). The ROC-AUC score of 0.89 further confirms the model's discriminative ability, indicating that it can effectively differentiate between classes. Additionally, Fig. 9 presents the decision boundary curve obtained from the model, visualizing how the random forest classifier separates the two classes. Overall, the random forest classifier demonstrates robust predictive performance, highlighting its potential utility in landslide risk assessment and early warning systems.

3.7 K-nearest neighbour

The K-nearest neighbour (KNN) classifier developed for predicting landslide occurrences based on rainfall intensity and duration has achieved the highest accuracy among all the models tested, with an impressive overall accuracy of 82.47%. This performance indicates that the KNN model effectively distinguishes between landslide and non-landslide points. The precision of 0.84 signifies that when the model predicts a landslide, it is correct 84% of the time, while the recall of 0.83 indicates that it successfully identifies 83% of actual landslide instances, demonstrating strong performance in capturing relevant events. The F1-score of 0.83 reflects a balanced performance between precision and recall. The confusion matrix reveals that the model correctly classifies 68 true landslides (true positives) and 59 true non-landslides (true negatives) while misclassifying 14 actual landslides as non-landslides (false negatives) and predicting 13 non-landslide points as landslides (false positives). The ROC-AUC score of 0.90 further confirms the model's excellent discriminative ability. Figure 10 presents the decision boundary plot obtained from the KNN classifier, providing a visual representation of how the model separates the two classes. Additionally, accuracy varies with the number of neighbors (K), as detailed in Table 2. Notably, the accuracy peaks at K = 5 with 82.47%, while varying for other values, such as 79.22% for K = 1, K = 3, and K = 4, and 76.62% for K = 2. Overall, the KNN classifier demonstrates robust predictive performance, emphasizing its potential utility in landslide risk assessment and early warning systems.

Table 3 presents the performance of all the machine learning classifiers utilized in this study and Fig. 11 shows the ROC-AUC of all the Models used. For a model to function effectively as a disaster prediction tool, two key factors are critical: strong prediction performance and minimal false positive results. In this context, the SVM classifier performed well in minimizing false positives; however, its prediction capability was the weakest among the models tested. On the other hand, both the RF and KNN classifiers demonstrated better performance, achieving higher true positive results. Overall, the KNN classifier outperformed the others, striking a balance with fewer false positives and a strong true positive rate.

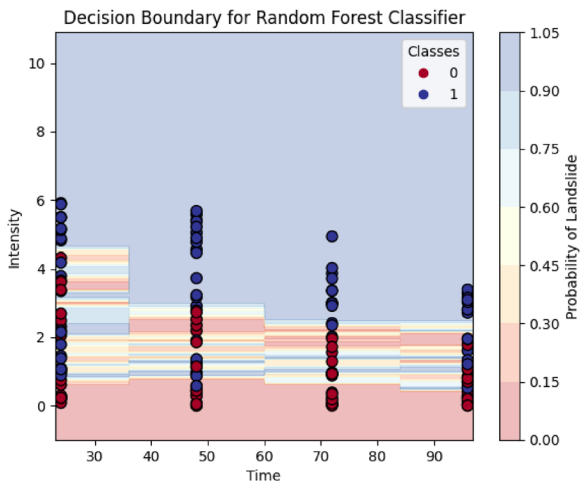


Fig. 9 Decision boundary for the random forest classifier

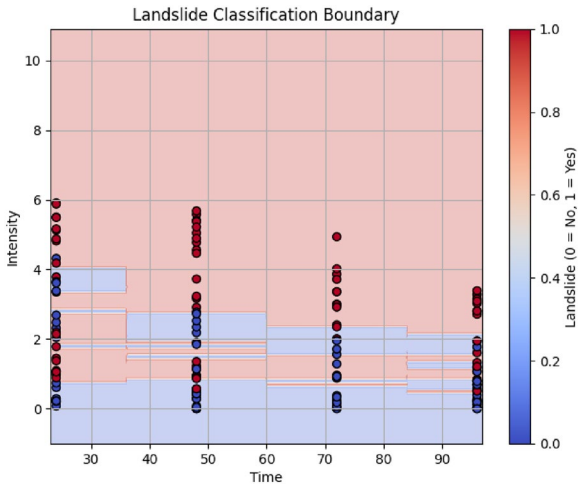


Fig. 10 Decision boundary for the KNN

Table 2 Change in accuracy with respect to change of K value

K value	Accuracy (%)
1	79.22
2	76.62
3	79.22
4	79.22
5	82.47
6	79.22
7	80.52
8	77.92
9	79.22
10	79.87

Table 3 Summary of the performance of all the machine learning classifiers used

Parameter	LR	GBM	SVM	RF	KNN
Accuracy	0.69	0.77	0.49	0.79	0.82
Precision	0.75	0.78	1.00	0.83	0.84
Recall	0.63	0.78	0.05	0.77	0.83
F1 Score	0.69	0.78	0.09	0.80	0.83
T^+	55	64	4	63	68
T^-	52	54	72	59	59
F^+	17	18	0	13	13
F^-	30	18	78	19	14

4 Conclusions

The present study explored the feasibility of creating a rainfall threshold model through various mathematical and statistical methods. This model is envisioned as a surrogate decision-making tool that can be integrated into a rain gauge-based device, functioning as an early warning system for landslides triggered by rainfall. The ability to predict landslides based on rainfall thresholds could significantly enhance disaster preparedness and response efforts. The data utilized in this research pertains to Kerala, India, with an assumption that the soil characteristics across the Western Ghats region are relatively uniform. This assumption underlines that the early warning system developed is site-specific and tailored to the unique environmental conditions and rainfall patterns of the region. This localized approach ensures that the system is relevant and effective for the target area. The present study yields several

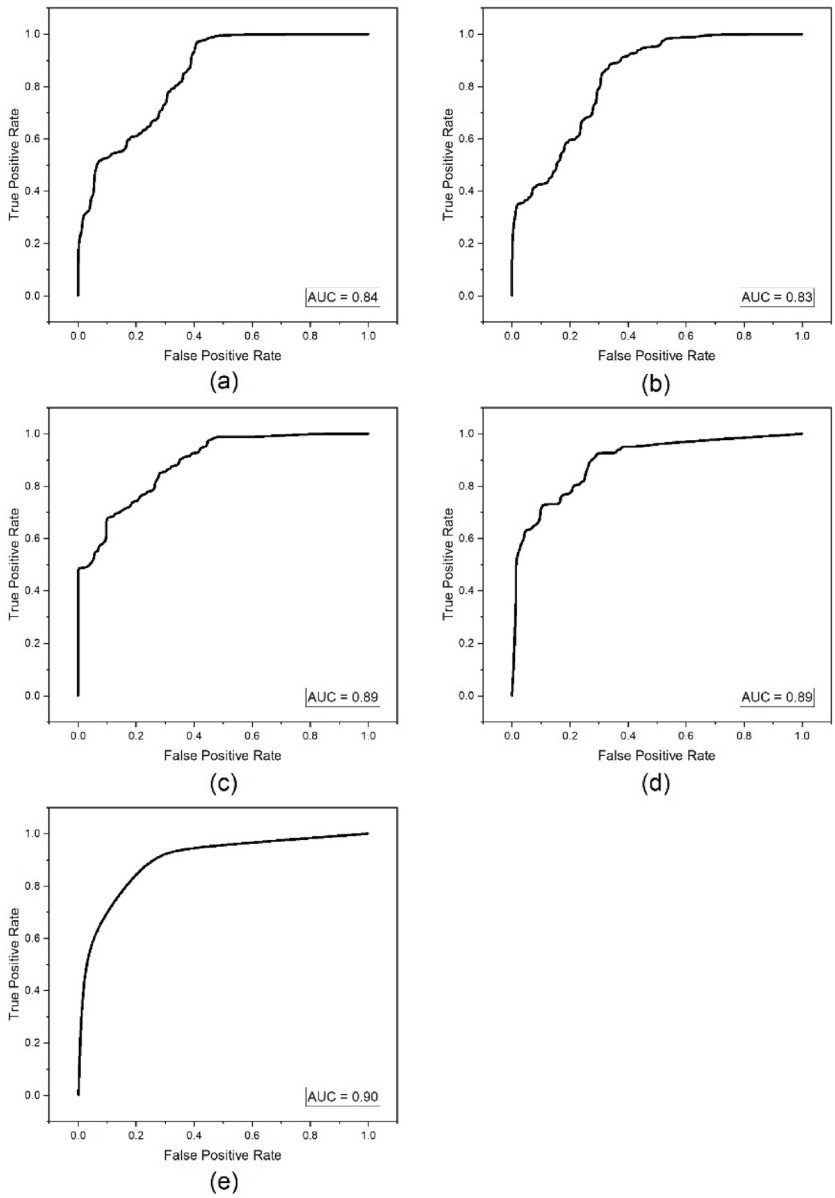


Fig. 11 ROC-AUC of **a** LR, **b** SVM, **c** GBM, **d** RF, and **e** KN

important findings regarding the development of an effective early warning system for rainfall-induced landslides, as summarized below.

1. The K-nearest neighbors (KNN) model exhibited the best performance metrics among the various models tested. However, it is crucial to recognize that the efficacy of any model depends on the specific characteristics of the data used. Hence, it is essential to filter the data appropriately and experiment with multiple models to determine which one yields the best results for the given dataset. This iterative process enhances the robustness of the predictive model.
2. The KNN-trained model can be saved and downloaded as a pickle file, a Python executable format that facilitates easy integration into real-time monitoring devices. This allows for seamless implementation in systems that measure rainfall, ensuring that the model can be utilized effectively in practice for timely alerts and risk assessments.
3. A larger dataset is likely to produce more accurate results, highlighting the importance of data quantity in model training. The study acknowledges that rainfall data is typically collected on a daily basis in India, as this traditional method persists in archival records. However, the limited availability of real-time monitoring devices, such as tipping bucket rain gauges, restricts the ability to capture precise rainfall duration and intensity. This gap indicates a need

for investment in modern monitoring technology to improve data accuracy and, consequently, the effectiveness of the early warning system.

Overall, these conclusions underscore the potential of integrating advanced modeling techniques into practical applications for disaster risk reduction, ultimately aiming to enhance safety and resilience in landslide-prone regions, which supports the Sustainable Development Goals 11 of the United Nations.

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Author contributions V.M. wrote the manuscript, conducted the analysis, and performed data collection. S.K. edited the manuscript and supervised the work.

Data availability All data used in this study are available from the corresponding author upon reasonable request. Additionally, the dataset utilized for the present study has been provided as supplementary material.

Declarations

Competing interests The authors declare no competing interests.

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